4607 - The Enigma of the Recurrent nova T Coronae Borealis - A ToO

Cycle: 3, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

Folder	*Observation*	*Label*	*Observing Template*	*Science Target*
TCrb-Visit1				
	1	TCrB-OnSource	MIRI Medium Resolution Spectroscopy	(1) V-T-CrB
	2	TCrb-BKGRND	MIRI Medium Resolution Spectroscopy	(2) V-T-CrB-Background
TCrb-Visit2				
	3	TCrB-OnSource	MIRI Medium Resolution Spectroscopy	(1) V-T-CrB
	4	TCrb-BKGRND	MIRI Medium Resolution Spectroscopy	(2) V-T-CrB-Background
TCrb-Visit3				
	5	TCrB-OnSource	MIRI Medium Resolution Spectroscopy	(1) V-T-CrB
	6	TCrb-BKGRND	MIRI Medium Resolution Spectroscopy	(2) V-T-CrB-Background
TCrb-Visit4				
	7	TCrB-OnSource	MIRI Medium Resolution Spectroscopy	(1) V-T-CrB
	8	TCrb-BKGRND	MIRI Medium Resolution Spectroscopy	(2) V-T-CrB-Background
TCrb-Visit5				
	9	TCrB-OnSource	MIRI Medium Resolution Spectroscopy	(1) V-T-CrB
	10	TCrb-BKGRND	MIRI Medium Resolution Spectroscopy	(2) V-T-CrB-Background

ABSTRACT

One of the most famous and brightest novae, the recurrent nova (RN) T Coronae Borealis, is expected to erupt during the period 2025.5 +/- 1.3. With only two recorded eruptions in 1866 and 1946, it has never been studied in the IR. T CrB declines in visual brightness very quickly after the nova explodes. During this decline towards quiescence, important and profound physical changes occur in the ejecta; e.g., the line profiles show rapid changes and coronal lines appear. Using the MRS IFU, we will observe and analyze the evolutions of: (1) shocks propagating through the ejecta, (2) the changes in velocity profiles of the emission lines, (3) the coronal line emission, deriving elemental abundances from the H, He and coronal lines, (4) dust features in the SED, and (5) ejecta-wind interactions and the possibility of discovering a light echo in the surrounding circumstellar environs. JWST observations will provide unprecedented insight into details of the nova's expanding fireball. They will have an invaluable impact on understanding symbiotic RNe, systems potentially Type 1a SN progenitors as their white dwarfs (WD) seem to exhibit a net accretion of material (i.e., from the Red Giant onto the WD minus that ejected in the RN explosion). This non-disruptive ToO will yield a thorough documentation of all phases (5 visits) of post-eruption evolution. JWST will contribute significantly to and complement a community-wide panchromatic observational campaign to study this extraordinary event. The T Crb event will further our understanding of RNe, yielding insight into Type Ia SN, which are key to determining cosmic large-scale structure.

OBSERVING DESCRIPTION

This is a target of opportunity (ToO) program as the observations are linked to an event that may occur at an unknown time. ToO targets include objects that can be identified in advance but which undergo unpredictable changes, as well as objects that can only be identified in advance as a class (e.g., T Crb is a recurrent novae). This proposal is a Non-disruptive ToO: A ToO for which the activation time is greater than 14 days. Special time constraints are selected to schedule the potential observations during peroids where the target is the the MAZ Safe Zone at the scientifically relevant cadence. The exact time and which window will be selected dependent on trigger, when the recurrent nova goes into outburst as communicated by various transient event brokers (ASSAN, Astronomer's Telegram [ATels], CBAT, etc.) The Principal Investigator (PI) or designated alternate will initiate activation of the Target of Opportunity (ToO) proposal by submitting an activation request, and then, after submitting the request, the PI (or alternate) will contact their Program Coordinator (PC) and verify that the activation request has been received by the Institute. We propose to observe T Crb with MRS in all bands and channels, on 5 temporally seperate synoptic epochs to follow the dynamic evolution of the eruption. Complete execution of the program may require observations scheduling over the nominal cycle boundary after initial activation, so long-term status is requested. Estimates of the probability of occurrence during Cy3 is 80%.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	V-T-CrB	RA: 15 59 30.1622 (239.8756758d) Dec: +25 55 12.61 (25.92017d) Equinox: J2000	Proper Motion RA: -4.461 mas/yr Proper Motion Dec: 12.016 mas/yr Parallax: 0.0011" Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. *Category=Star* *Description=[Recurrent novae, Stellar winds, Symbiotic binary stars]*			
(2)	V-T-CrB-Background	RA: 15 59 26.0100 (239.8583750d) Dec: +25 56 59.50 (25.94986d) Equinox: J2000	Proper Motion RA: -4.461 mas/yr Proper Motion Dec: 12.016 mas/yr Parallax: 0.0011" Epoch of Position: 2000	
	Comments: Background position for MRS selected after inspection of field near target, nothing evident in either WISE or 2MASS toward the position after inspection in Aladin in the APT *Category=Star* *Description=[Recurrent novae, Stellar winds, Symbiotic binary stars]*			

Observation

Proposal 4607, Observation 1: TCrB-OnSource Tue Jun 04 00:03:19 GMT 2024

Diagnostic Status: Warning

Observing Template: MIRI Medium Resolution Spectroscopy

Background Observations:[TCrb-BKGRND (Obs 2)]

Comments: This is a target of opportunity (ToO) program as the observations are linked to an event that may occur at an unknown time. ToO targets include objects that can be identified in advance but which undergo unpredictable changes, as well as objects that can only be identified in advance as a class (e.g., T Crb is a recurrent novae). This proposal is a Non-disruptive ToO: A ToO for which the activation time is greater than 14 days. The between dates (3) special time constraints are selected to schedule the potential observation during peroids where the target is the the MAZ Safe Zone. The exact time and which window will be selected dependent on trigger, when the recurrent nova goes into outburst.

Diagnostics

(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	V-T-CrB	RA: 15 59 30.1622 (239.8756758d) Dec: +25 55 12.61 (25.92017d) Equinox: J2000	Proper Motion RA: -4.461 mas/yr Proper Motion Dec: 12.016 mas/yr Parallax: 0.0011" Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=Star
Description=[Recurrent novae, Stellar winds, Symbiotic binary stars]

Acquisition

#	Target
1	NONE

Template

AcqFilter	Primary Channel	Simultaneous Imaging	Imager Subarray	Grating Wheel Direction
	All MRS	NO	FULL	NEUTRAL

Dithers

#	Dither Type	Optimized For	Direction
1	4-Point	POINT SOURCE	NEGATIVE

Spectral Elements

#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	SHORT(A)	MRSLONG		FASTR1	6	1	1	Dither 1	4	4	66.601	168795
1	SHORT(A)	MRSSHORT		FASTR1	6	1	1	Dither 1	4	4	66.601	168795
2	MEDIUM(B)	MRSLONG		FASTR1	6	1	1	Dither 1	4	4	66.601	168795
2	MEDIUM(B)	MRSSHORT		FASTR1	6	1	1	Dither 1	4	4	66.601	168795
3	LONG(C)	MRSLONG		FASTR1	6	1	1	Dither 1	4	4	66.601	168795
3	LONG(C)	MRSSHORT		FASTR1	6	1	1	Dither 1	4	4	66.601	168795

Special Requirements	
	Target Of Opportunity Response Time 14 Days, Carry-Over Sequence Observations 1, 2, Non-interruptible Sequence Observations 1, 3 within 15 Days Sequence Observations 1, 5 within 70 Days Sequence Observations 1, 7 within 160 Days Sequence Observations 1, 9 within 200 Days

Observation

Proposal 4607, Observation 2: TCrb-BKGRND

Tue Jun 04 00:03:19 GMT 2024

Diagnostic Status: Warning

Observing Template: MIRI Medium Resolution Spectroscopy

Background Observation For: [TCrB-OnSource (Obs 1)]

Diagnostics

(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(2)	V-T-CrB-Background	RA: 15 59 26.0100 (239.8583750d)	Proper Motion RA: -4.461 mas/yr	
		Dec: +25 56 59.50 (25.94986d)	Proper Motion Dec: 12.016 mas/yr	
		Equinox: J2000	Parallax: 0.0011"	
			Epoch of Position: 2000	

Comments: Background position for MRS selected after inspection of field near target, nothing evident in either WISE or 2MASS toward the position after inspection in Aladin in the APT
Category=Star
Description=[Recurrent novae, Stellar winds, Symbiotic binary stars]

Acquisition

#	Target
1	NONE

Template

AcqFilter	Primary Channel	Simultaneous Imaging	Imager Subarray	Grating Wheel Direction
	All MRS	NO	FULL	NEUTRAL

Dithers

#	Dither Type	Optimized For	Direction
1	4-Point	POINT SOURCE	NEGATIVE

Spectral Elements

#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	SHORT(A)	MRSLONG		FASTR1	6	1	1	Dither 1	4	4	66.601	
1	SHORT(A)	MRSSHORT		FASTR1	6	1	1	Dither 1	4	4	66.601	
2	MEDIUM(B)	MRSLONG		FASTR1	6	1	1	Dither 1	4	4	66.601	
2	MEDIUM(B)	MRSSHORT		FASTR1	6	1	1	Dither 1	4	4	66.601	
3	LONG(C)	MRSLONG		FASTR1	6	1	1	Dither 1	4	4	66.601	
3	LONG(C)	MRSSHORT		FASTR1	6	1	1	Dither 1	4	4	66.601	

Special Requirements	Sequence Observations 1, 2, Non-interruptible

Observation

Proposal 4607, Observation 3: TCrB-OnSource

Tue Jun 04 00:03:19 GMT 2024

Diagnostic Status: Warning

Observing Template: MIRI Medium Resolution Spectroscopy

Background Observations:[TCrb-BKGRND (Obs 4)]

Comments: This is a target of opportunity (ToO) program as the observations are linked to an event that may occur at an unknown time. ToO targets include objects that can be identified in advance but which undergo unpredictable changes, as well as objects that can only be identified in advance as a class (e.g., T Crb is a recurrent novae). This proposal is a Non-disruptive ToO: A ToO for which the activation time is greater than 14 days. The between dates (3) special time constraints are selected to schedule the potential observation during peroids where the target is the the MAZ Safe Zone. The exact time and which window will be selected dependent on trigger, when the recurrent nova goes into outburst.

Diagnostics

(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	V-T-CrB	RA: 15 59 30.1622 (239.8756758d)	Proper Motion RA: -4.461 mas/yr	
		Dec: +25 55 12.61 (25.92017d)	Proper Motion Dec: 12.016 mas/yr	
		Equinox: J2000	Parallax: 0.0011"	
			Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=Star
Description=[Recurrent novae, Stellar winds, Symbiotic binary stars]

Acquisition

#	Target
1	NONE

Template

AcqFilter	Primary Channel	Simultaneous Imaging	Imager Subarray	Grating Wheel Direction
	All MRS	NO	FULL	NEUTRAL

Dithers

#	Dither Type	Optimized For	Direction
1	4-Point	POINT SOURCE	NEGATIVE

Spectral Elements

#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	SHORT(A)	MRSLONG		FASTR1	6	1	1	Dither 1	4	4	66.601	168795
1	SHORT(A)	MRSSHORT		FASTR1	6	1	1	Dither 1	4	4	66.601	168795
2	MEDIUM(B)	MRSLONG		FASTR1	6	1	1	Dither 1	4	4	66.601	168795
2	MEDIUM(B)	MRSSHORT		FASTR1	6	1	1	Dither 1	4	4	66.601	168795
3	LONG(C)	MRSLONG		FASTR1	6	1	1	Dither 1	4	4	66.601	168795
3	LONG(C)	MRSSHORT		FASTR1	6	1	1	Dither 1	4	4	66.601	168795

Special Requirements	
	Sequence Observations 1, 3 within 15 Days Sequence Observations 3, 4, Non-interruptible

Observation

Proposal 4607, Observation 4: TCrb-BKGRND

Tue Jun 04 00:03:19 GMT 2024

Diagnostic Status: Warning

Observing Template: MIRI Medium Resolution Spectroscopy

Background Observation For: [TCrB-OnSource (Obs 3)]

Diagnostics

(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(2)	V-T-CrB-Background	RA: 15 59 26.0100 (239.8583750d)	Proper Motion RA: -4.461 mas/yr	
		Dec: +25 56 59.50 (25.94986d)	Proper Motion Dec: 12.016 mas/yr	
		Equinox: J2000	Parallax: 0.0011"	
			Epoch of Position: 2000	

Comments: Background position for MRS selected after inspection of field near target, nothing evident in either WISE or 2MASS toward the position after inspection in Aladin in the APT
Category=Star
Description=[Recurrent novae, Stellar winds, Symbiotic binary stars]

Acquisition

#	Target
1	NONE

Template

AcqFilter	Primary Channel	Simultaneous Imaging	Imager Subarray	Grating Wheel Direction
	All MRS	NO	FULL	NEUTRAL

Dithers

#	Dither Type	Optimized For	Direction
1	4-Point	POINT SOURCE	NEGATIVE

Spectral Elements

#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	SHORT(A)	MRSLONG		FASTR1	6	1	1	Dither 1	4	4	66.601	
1	SHORT(A)	MRSSHORT		FASTR1	6	1	1	Dither 1	4	4	66.601	
2	MEDIUM(B)	MRSLONG		FASTR1	6	1	1	Dither 1	4	4	66.601	
2	MEDIUM(B)	MRSSHORT		FASTR1	6	1	1	Dither 1	4	4	66.601	
3	LONG(C)	MRSLONG		FASTR1	6	1	1	Dither 1	4	4	66.601	
3	LONG(C)	MRSSHORT		FASTR1	6	1	1	Dither 1	4	4	66.601	

Special Requirements	Sequence Observations 3, 4, Non-interruptible

Observation

Proposal 4607, Observation 5: TCrB-OnSource Tue Jun 04 00:03:19 GMT 2024

Diagnostic Status: Warning

Observing Template: MIRI Medium Resolution Spectroscopy

Background Observations:[TCrb-BKGRND (Obs 6)]

Comments: This is a target of opportunity (ToO) program as the observations are linked to an event that may occur at an unknown time. ToO targets include objects that can be identified in advance but which undergo unpredictable changes, as well as objects that can only be identified in advance as a class (e.g., T Crb is a recurrent novae). This proposal is a Non-disruptive ToO: A ToO for which the activation time is greater than 14 days. The between dates (3) special time constraints are selected to schedule the potential observation during peroids where the target is the the MAZ Safe Zone. The exact time and which window will be selected dependent on trigger, when the recurrent nova goes into outburst.

Diagnostics

(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	V-T-CrB	RA: 15 59 30.1622 (239.8756758d)	Proper Motion RA: -4.461 mas/yr	
		Dec: +25 55 12.61 (25.92017d)	Proper Motion Dec: 12.016 mas/yr	
		Equinox: J2000	Parallax: 0.0011"	
			Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=Star
Description=[Recurrent novae, Stellar winds, Symbiotic binary stars]

Acquisition

#	Target
1	NONE

Template

AcqFilter	Primary Channel	Simultaneous Imaging	Imager Subarray	Grating Wheel Direction
	All MRS	NO	FULL	NEUTRAL

Dithers

#	Dither Type	Optimized For	Direction
1	4-Point	POINT SOURCE	NEGATIVE

Spectral Elements

#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	SHORT(A)	MRSLONG		FASTR1	6	1	1	Dither 1	4	4	66.601	168795
1	SHORT(A)	MRSSHORT		FASTR1	6	1	1	Dither 1	4	4	66.601	168795
2	MEDIUM(B)	MRSLONG		FASTR1	6	1	1	Dither 1	4	4	66.601	168795
2	MEDIUM(B)	MRSSHORT		FASTR1	6	1	1	Dither 1	4	4	66.601	168795
3	LONG(C)	MRSLONG		FASTR1	6	1	1	Dither 1	4	4	66.601	168795
3	LONG(C)	MRSSHORT		FASTR1	6	1	1	Dither 1	4	4	66.601	168795

Special Requirements	
	Sequence Observations 1, 5 within 70 Days Sequence Observations 5, 6, Non-interruptible

Observation

Proposal 4607, Observation 6: TCrb-BKGRND

Tue Jun 04 00:03:19 GMT 2024

Diagnostic Status: Warning

Observing Template: MIRI Medium Resolution Spectroscopy

Background Observation For: [TCrB-OnSource (Obs 5)]

Diagnostics

(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(2)	V-T-CrB-Background	RA: 15 59 26.0100 (239.8583750d) Dec: +25 56 59.50 (25.94986d) Equinox: J2000	Proper Motion RA: -4.461 mas/yr Proper Motion Dec: 12.016 mas/yr Parallax: 0.0011" Epoch of Position: 2000	

Comments: Background position for MRS selected after inspection of field near target, nothing evident in either WISE or 2MASS toward the position after inspection in Aladin in the APT
Category=Star
Description=[Recurrent novae, Stellar winds, Symbiotic binary stars]

Acquisition

#	Target
1	NONE

Template

AcqFilter	Primary Channel	Simultaneous Imaging	Imager Subarray	Grating Wheel Direction
	All MRS	NO	FULL	NEUTRAL

Dithers

#	Dither Type	Optimized For	Direction
1	4-Point	POINT SOURCE	NEGATIVE

Spectral Elements

#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	SHORT(A)	MRSLONG		FASTR1	6	1	1	Dither 1	4	4	66.601	
1	SHORT(A)	MRSSHORT		FASTR1	6	1	1	Dither 1	4	4	66.601	
2	MEDIUM(B)	MRSLONG		FASTR1	6	1	1	Dither 1	4	4	66.601	
2	MEDIUM(B)	MRSSHORT		FASTR1	6	1	1	Dither 1	4	4	66.601	
3	LONG(C)	MRSLONG		FASTR1	6	1	1	Dither 1	4	4	66.601	
3	LONG(C)	MRSSHORT		FASTR1	6	1	1	Dither 1	4	4	66.601	

Special Requirements	
	Sequence Observations 5, 6, Non-interruptible

Observation

Proposal 4607, Observation 7: TCrB-OnSource Tue Jun 04 00:03:19 GMT 2024

Diagnostic Status: Warning

Observing Template: MIRI Medium Resolution Spectroscopy

Background Observations:[TCrb-BKGRND (Obs 8)]

Comments: This is a target of opportunity (ToO) program as the observations are linked to an event that may occur at an unknown time. ToO targets include objects that can be identified in advance but which undergo unpredictable changes, as well as objects that can only be identified in advance as a class (e.g., T Crb is a recurrent novae). This proposal is a Non-disruptive ToO: A ToO for which the activation time is greater than 14 days. The between dates (3) special time constraints are selected to schedule the potential observation during peroids where the target is the the MAZ Safe Zone. The exact time and which window will be selected dependent on trigger, when the recurrent nova goes into outburst.

Diagnostics

(Visit 7:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	V-T-CrB	RA: 15 59 30.1622 (239.8756758d)	Proper Motion RA: -4.461 mas/yr	
		Dec: +25 55 12.61 (25.92017d)	Proper Motion Dec: 12.016 mas/yr	
		Equinox: J2000	Parallax: 0.0011"	
			Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=Star
Description=[Recurrent novae, Stellar winds, Symbiotic binary stars]

Acquisition

#	Target
1	NONE

Template

AcqFilter	Primary Channel	Simultaneous Imaging	Imager Subarray	Grating Wheel Direction
	All MRS	NO	FULL	NEUTRAL

Dithers

#	Dither Type	Optimized For	Direction
1	4-Point	POINT SOURCE	NEGATIVE

Spectral Elements

#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	SHORT(A)	MRSLONG		FASTR1	6	1	1	Dither 1	4	4	66.601	168795
1	SHORT(A)	MRSSHORT		FASTR1	6	1	1	Dither 1	4	4	66.601	168795
2	MEDIUM(B)	MRSLONG		FASTR1	6	1	1	Dither 1	4	4	66.601	168795
2	MEDIUM(B)	MRSSHORT		FASTR1	6	1	1	Dither 1	4	4	66.601	168795
3	LONG(C)	MRSLONG		FASTR1	6	1	1	Dither 1	4	4	66.601	168795
3	LONG(C)	MRSSHORT		FASTR1	6	1	1	Dither 1	4	4	66.601	168795

Proposal 4607 - Observation 7 - The Enigma of the Recurrent nova T Coronae Borealis - A ToO

Special Requirements	
	Sequence Observations 1, 7 within 160 Days Sequence Observations 7, 8, Non-interruptible

Observation

Proposal 4607, Observation 8: TCrb-BKGRND

Tue Jun 04 00:03:19 GMT 2024

Diagnostic Status: Warning

Observing Template: MIRI Medium Resolution Spectroscopy

Background Observation For: [TCrB-OnSource (Obs 7)]

Diagnostics

(Visit 8:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(2)	V-T-CrB-Background	RA: 15 59 26.0100 (239.8583750d)	Proper Motion RA: -4.461 mas/yr	
		Dec: +25 56 59.50 (25.94986d)	Proper Motion Dec: 12.016 mas/yr	
		Equinox: J2000	Parallax: 0.0011"	
			Epoch of Position: 2000	

Comments: Background position for MRS selected after inspection of field near target, nothing evident in either WISE or 2MASS toward the position after inspection in Aladin in the APT
Category=Star
Description=[Recurrent novae, Stellar winds, Symbiotic binary stars]

Acquisition

#	Target
1	NONE

Template

AcqFilter	Primary Channel	Simultaneous Imaging	Imager Subarray	Grating Wheel Direction
	All MRS	NO	FULL	NEUTRAL

Dithers

#	Dither Type	Optimized For	Direction
1	4-Point	POINT SOURCE	NEGATIVE

Spectral Elements

#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	SHORT(A)	MRSLONG		FASTR1	6	1	1	Dither 1	4	4	66.601	
1	SHORT(A)	MRSSHORT		FASTR1	6	1	1	Dither 1	4	4	66.601	
2	MEDIUM(B)	MRSLONG		FASTR1	6	1	1	Dither 1	4	4	66.601	
2	MEDIUM(B)	MRSSHORT		FASTR1	6	1	1	Dither 1	4	4	66.601	
3	LONG(C)	MRSLONG		FASTR1	6	1	1	Dither 1	4	4	66.601	
3	LONG(C)	MRSSHORT		FASTR1	6	1	1	Dither 1	4	4	66.601	

Special Requirements	Sequence Observations 7, 8, Non-interruptible

Observation

Proposal 4607, Observation 9: TCrB-OnSource

Tue Jun 04 00:03:19 GMT 2024

Diagnostic Status: Warning

Observing Template: MIRI Medium Resolution Spectroscopy

Background Observations:[TCrb-BKGRND (Obs 10)]

Comments: This is a target of opportunity (ToO) program as the observations are linked to an event that may occur at an unknown time. ToO targets include objects that can be identified in advance but which undergo unpredictable changes, as well as objects that can only be identified in advance as a class (e.g., T Crb is a recurrent novae). This proposal is a Non-disruptive ToO: A ToO for which the activation time is greater than 14 days. The between dates (3) special time constraints are selected to schedule the potential observation during peroids where the target is the the MAZ Safe Zone. The exact time and which window will be selected dependent on trigger, when the recurrent nova goes into outburst.

Diagnostics

(Visit 9:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	V-T-CrB	RA: 15 59 30.1622 (239.8756758d) Dec: +25 55 12.61 (25.92017d) Equinox: J2000	Proper Motion RA: -4.461 mas/yr Proper Motion Dec: 12.016 mas/yr Parallax: 0.0011" Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=Star
Description=[Recurrent novae, Stellar winds, Symbiotic binary stars]

Acquisition

#	Target
1	NONE

Template

AcqFilter	Primary Channel	Simultaneous Imaging	Imager Subarray	Grating Wheel Direction
	All MRS	NO	FULL	NEUTRAL

Dithers

#	Dither Type	Optimized For	Direction
1	4-Point	POINT SOURCE	NEGATIVE

Spectral Elements

#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	SHORT(A)	MRSLONG		FASTR1	6	1	1	Dither 1	4	4	66.601	168795
1	SHORT(A)	MRSSHORT		FASTR1	6	1	1	Dither 1	4	4	66.601	168795
2	MEDIUM(B)	MRSLONG		FASTR1	6	1	1	Dither 1	4	4	66.601	168795
2	MEDIUM(B)	MRSSHORT		FASTR1	6	1	1	Dither 1	4	4	66.601	168795
3	LONG(C)	MRSLONG		FASTR1	6	1	1	Dither 1	4	4	66.601	168795
3	LONG(C)	MRSSHORT		FASTR1	6	1	1	Dither 1	4	4	66.601	168795

Special Requirements	
	Sequence Observations 1, 9 within 200 Days Sequence Observations 9, 10, Non-interruptible

Observation

Proposal 4607, Observation 10: TCrb-BKGRND Tue Jun 04 00:03:19 GMT 2024

Diagnostic Status: Warning

Observing Template: MIRI Medium Resolution Spectroscopy

Background Observation For: [TCrB-OnSource (Obs 9)]

Diagnostics

(Visit 10:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(2)	V-T-CrB-Background	RA: 15 59 26.0100 (239.8583750d)	Proper Motion RA: -4.461 mas/yr	
		Dec: +25 56 59.50 (25.94986d)	Proper Motion Dec: 12.016 mas/yr	
		Equinox: J2000	Parallax: 0.0011"	
			Epoch of Position: 2000	

Comments: Background position for MRS selected after inspection of field near target, nothing evident in either WISE or 2MASS toward the position after inspection in Aladin in the APT
Category=Star
Description=[Recurrent novae, Stellar winds, Symbiotic binary stars]

Acquisition

#	Target
1	NONE

Template

AcqFilter	Primary Channel	Simultaneous Imaging	Imager Subarray	Grating Wheel Direction
	All MRS	NO	FULL	NEUTRAL

Dithers

#	Dither Type	Optimized For	Direction
1	4-Point	POINT SOURCE	NEGATIVE

Spectral Elements

#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	SHORT(A)	MRSLONG		FASTR1	6	1	1	Dither 1	4	4	66.601	
1	SHORT(A)	MRSSHORT		FASTR1	6	1	1	Dither 1	4	4	66.601	
2	MEDIUM(B)	MRSLONG		FASTR1	6	1	1	Dither 1	4	4	66.601	
2	MEDIUM(B)	MRSSHORT		FASTR1	6	1	1	Dither 1	4	4	66.601	
3	LONG(C)	MRSLONG		FASTR1	6	1	1	Dither 1	4	4	66.601	
3	LONG(C)	MRSSHORT		FASTR1	6	1	1	Dither 1	4	4	66.601	

Special Requirements	Sequence Observations 9, 10, Non-interruptible